

Fluid Substitution and Pre-Stack Seismic Modeling

AVO-Based Fluid Discrimination in a Shaly-Sand
Clastic Reservoir

A Rock-Physics and Quantitative Interpretation Study

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Abstract

This study applies a rock-physics and quantitative-interpretation workflow to a synthetic, shaly-sand Tertiary clastic reservoir in order to predict the seismic amplitude-versus-offset (AVO) response to changes in pore fluid. Gassmann fluid substitution is used to transform an in-situ, brine-saturated reservoir sand into oil- and gas-charged equivalents, from which a V_p/V_s -versus-acoustic-impedance rock physics template is constructed and calibrated against synthetic well-log and core data. Pre-stack synthetic angle gathers, generated with a three-term linearized reflectivity approximation and a zero-phase wavelet, are used to classify the resulting AVO anomalies. The brine case yields a Class I response, the oil case a notably subtle, near-zero-intercept Class II/IIp response that requires full-angle gradient analysis to resolve, and the gas case an unambiguous Class III bright-spot response. The workflow – and every quantitative result reported below – is fully reproducible from the accompanying source code.

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1 Introduction and Objectives

Discriminating pore fluid from lithology is one of the central problems in quantitative seismic interpretation. A change in pore fluid alters the bulk modulus of a reservoir rock while leaving its shear modulus essentially unchanged, and this asymmetry is what makes amplitude-versus-offset (AVO) analysis diagnostic: the intercept and gradient of the pre-stack reflection amplitude, taken together, can separate a fluid effect from a purely lithological one in a way that a post-stack amplitude alone cannot.

This study works through that diagnostic chain end to end for a single, representative reservoir interval:

- (i) build a physically consistent well-log model of a shaly-sand reservoir;
- (ii) use Gassmann fluid substitution to predict how that sand’s elastic properties change when it is charged with oil or gas instead of brine;
- (iii) calibrate a rock physics template that generalizes this fluid sensitivity across a range of porosity, for lithofacies and fluid-type discrimination; and
- (iv) forward-model the pre-stack seismic response of each fluid case, and classify the resulting AVO anomaly.

No proprietary or measured field dataset is used at any stage. The well-log model is a synthetic “type well,” built from standard, cited empirical and petrophysical relations and calibrated to parameter ranges that are typical of a shallow, high-porosity, under-compacted Tertiary clastic section (broadly representative of North Sea-type basins). This choice keeps every input transparent and reproducible, and lets the workflow focus on the physics of fluid substitution and AVO response rather than on well-specific data conditioning.

2 Geological Setting and Synthetic Well Data

The modelled section spans 2000–2200 m true vertical depth, representative of a normally pressured, moderately young (Tertiary) clastic basin. A single, 16 m-thick, high-net-to-gross reservoir sand is present from 2140–2156 m, embedded in a shale-dominated background that also contains two thin, non-reservoir silty streaks included for log realism but excluded from the fluid-substitution and AVO analysis.

Shale volume (V_{sh}) follows a smooth, log-resolution-limited transition into and out of the reservoir sand, superimposed on a slowly varying regional background and a band-limited noise texture that mimics real wireline log character. Porosity follows an Athy’s-law (Athy, 1930) exponential compaction trend, distinct for the sand and shale end-members and mixed according to V_{sh} :

$$\phi(z) = \phi_0 e^{-cz},$$

with $\phi_{0,\text{sand}} = 0.45$, $c_{\text{sand}} = 1.2 \times 10^{-4} \text{ m}^{-1}$, $\phi_{0,\text{shale}} = 0.55$, $c_{\text{shale}} = 3.5 \times 10^{-4} \text{ m}^{-1}$. This produces a clean reservoir-sand porosity of approximately 33–35%, consistent with the very high porosities documented in young, shallow, poorly consolidated Tertiary sands.

In-situ pressure and temperature follow simple linear gradients appropriate to a normally pressured basin: a hydrostatic pressure gradient of 10.0 MPa/km, and a geothermal gradient of 30 °C/km from an 8 °C surface (mudline) temperature. At the reservoir midpoint (2148 m) this gives $T \approx 72.4$ °C and $P \approx 21.5$ MPa.

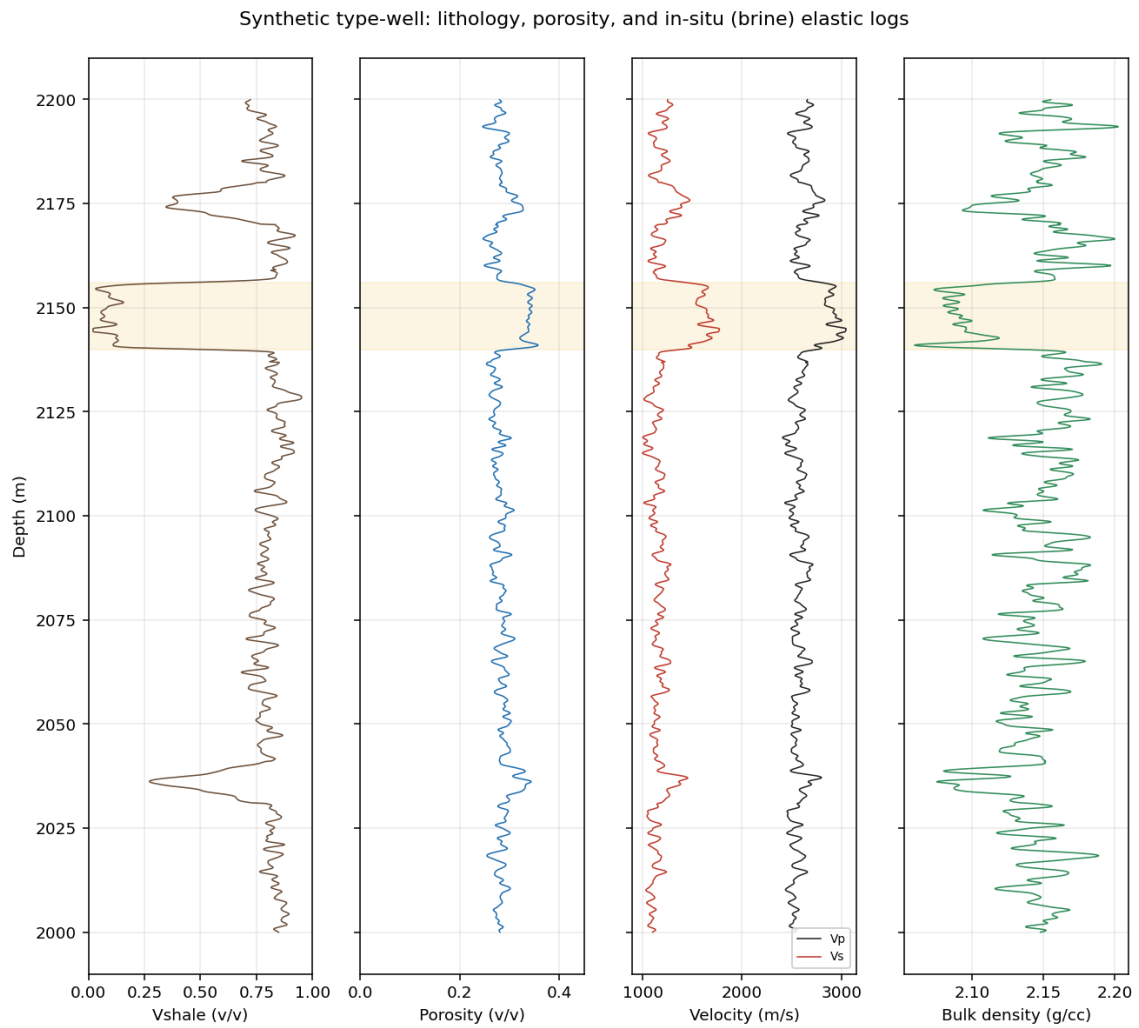


Figure 1: Synthetic type-well: lithology (V_{shale}), porosity, and in-situ (brine-saturated) V_p/V_s and bulk density logs. The reservoir interval (2140–2156 m) is highlighted.

3 Methodology

3.1 Mineral and Fluid Properties

Mineral moduli are computed as a Voigt-Reuss-Hill (Hill, 1952) average of quartz and clay end-members, weighted by V_{sh} :

$$M_{\text{VRH}} = \frac{1}{2} \left[(f_1 M_1 + f_2 M_2) + \left(\frac{f_1}{M_1} + \frac{f_2}{M_2} \right)^{-1} \right],$$

applied separately to the bulk (K) and shear (μ) moduli, using $K_{\text{qtz}} = 36.6$, $\mu_{\text{qtz}} = 45.0$ GPa, $\rho_{\text{qtz}} = 2650$ kg/m³ for quartz and $K_{\text{clay}} = 20.9$, $\mu_{\text{clay}} = 6.85$ GPa, $\rho_{\text{clay}} = 2580$ kg/m³ for clay (representative literature values).

Brine density and acoustic velocity are computed with the Batzle & Wang (1992) empirical relations (their eqs. 27b, 29), at 35,000 ppm NaCl salinity, using the open-source **bruges** rock-physics library’s implementation. Oil is modelled as a dead (gas-free), 32° API crude: density follows exactly from the API-gravity definition at stock-tank conditions, corrected to reservoir temperature and pressure with a simple linear thermal-expansion/compressibility relation, and combined with a representative reservoir-condition acoustic velocity (consistent with the range reported by Batzle & Wang for a dead oil of this gravity) to obtain the bulk modulus. Gas is modelled under an ideal-gas approximation, consistent with the density relation used internally by the brine/gas routines this project depends on: $\rho_g = MP/RT$ and an adiabatic bulk modulus $K_g = \gamma P$, with an effective heat-capacity ratio $\gamma = 1.25$ representative of a light hydrocarbon gas. A full live-oil/real-gas PVT treatment (requiring measured GOR, bubble point, and gas composition) was not attempted, since no such measured PVT report exists for this synthetic case; this simplification is noted explicitly here and in the accompanying source code.

The resulting fluid properties at reservoir conditions ($T = 72.4^\circ\text{C}$, $P = 21.5$ MPa) are summarized in Table 1.

Table 1: Pore-fluid properties at in-situ reservoir conditions.

Fluid	Density (kg/m ³)	Bulk modulus (GPa)	Basis
Brine (35 kppm NaCl)	1010	2.673	Batzle & Wang (1992)
Oil (32° API, dead)	840	1.577	Simplified dead-oil PVT
Gas ($\gamma_g = 0.65$)	141	0.0269	Ideal-gas approximation

3.2 Dry-Frame Rock Physics Model

Dry-frame bulk and shear moduli are estimated with the Krief et al. (1990) relation, which links the frame moduli to the mineral moduli through porosity alone:

$$\frac{K_{\text{dry}}}{K_{\text{min}}} = \frac{\mu_{\text{dry}}}{\mu_{\text{min}}} = (1 - \phi)^{A/(1-\phi)}, \quad A = 3.$$

This model was chosen over a contact-cement or Hertz-Mindlin granular model because it requires no additional core-calibrated cementation or coordination-number parameters, which is appropriate for a first-pass, log-driven workflow of this kind.

3.3 Gassmann Fluid Substitution

Gassmann's (1951) equation relates the saturated-rock bulk modulus K_{sat} to the dry-frame modulus K_{dry} , the mineral modulus K_{min} , the pore-fluid modulus K_{fl} , and porosity ϕ :

$$K_{\text{sat}} = K_{\text{dry}} + \frac{(1 - K_{\text{dry}}/K_{\text{min}})^2}{\phi/K_{\text{fl}} + (1 - \phi)/K_{\text{min}} - K_{\text{dry}}/K_{\text{min}}^2}.$$

The shear modulus is, by Gassmann's central assumption, unaffected by the pore fluid: $\mu_{\text{sat}} = \mu_{\text{dry}}$. Bulk density follows directly from a linear mixture of the mineral and fluid densities, $\rho = (1 - \phi)\rho_{\text{min}} + \phi\rho_{\text{fl}}$, so that substituting fluid A for fluid B at fixed porosity simply requires $\rho_{\text{new}} = \rho_{\text{old}} - \phi(\rho_{\text{fl,A}} - \rho_{\text{fl,B}})$.

Within the 2140–2156 m reservoir interval, the fully brine-saturated log (Section 4.1) is treated as the in-situ elastic state, and Gassmann's equation is applied twice – once with the oil modulus, once with the gas modulus – to predict the corresponding hydrocarbon-charged elastic logs, at fixed porosity and dry-frame moduli. As a self-consistency check, substituting brine back in for brine was verified to reproduce the original log exactly (to machine precision), confirming the forward/inverse formulation is implemented correctly.

3.4 Rock Physics Template Construction

A rock physics template (RPT) generalizes the single-interval substitution above across a full porosity range (5–40%), at fixed mineralogy and in-situ conditions, tracing out continuous brine-sand, oil-sand, gas-sand, and shale trends in V_p/V_s -versus-acoustic-impedance ($\text{AI} = V_p\rho$) space. To emulate a log-and-core calibrated template, a set of sparse synthetic "core plug" samples was generated along the sand trend with independent porosity scatter and measurement noise on V_p , V_s , and ρ .

3.5 Pre-Stack Synthetic Seismic Modeling

Reflectivity as a function of incidence angle θ is computed with the Shuey (1985) three-term approximation to the linearized Aki & Richards (1980) reflection coefficient:

$$R(\theta) = A + B \sin^2 \theta + C \sin^2 \theta \tan^2 \theta,$$

$$A = \frac{1}{2} \left(\frac{\Delta V_p}{\bar{V}_p} + \frac{\Delta \rho}{\bar{\rho}} \right), \quad B = \frac{1}{2} \frac{\Delta V_p}{\bar{V}_p} - 2 \left(\frac{\bar{V}_s}{\bar{V}_p} \right)^2 \left(\frac{\Delta \rho}{\bar{\rho}} + 2 \frac{\Delta V_s}{\bar{V}_s} \right), \quad C = \frac{1}{2} \frac{\Delta V_p}{\bar{V}_p},$$

where overbars denote the average and Δ the contrast of a property across the interface. This approximation is accurate to roughly 30–35°, which bounds the 0–35° angle range modelled here.

Synthetic angle gathers are built by placing spike reflectivities – scaled by $R(\theta)$ – at the two-way travel times of the top- and base-of-reservoir interfaces, and convolving with a zero-phase Ricker wavelet of dominant frequency 30 Hz:

$$w(t) = [1 - 2(\pi ft)^2] \exp [-(\pi ft)^2].$$

A constant overburden velocity (2400 m/s) is used only to position the two reflectors in time; it does not enter the reflectivity modelling itself, which uses the layer elastic properties directly. Near (0–12°), mid (12–24°), and far (24–35°) angle stacks are formed by simple averaging within each range, and each fluid case's top-of-reservoir response is classified using the two-term intercept/gradient scheme of Rutherford & Williams (1989) and Castagna & Swan (1997).

4 Results

4.1 Well Log and Depth Trend Analysis

Figure 1 shows the synthetic well log suite. The reservoir sand is clearly expressed as a low- V_{sh} , high-porosity interval with a corresponding velocity and density response relative to the shale background. Figure 2 compares the depth trend of acoustic impedance and V_p/V_s for the in-situ brine case against the Gassmann-substituted oil and gas cases within the reservoir: both attributes decrease systematically as the pore fluid becomes lighter, with the largest step occurring at the gas substitution.

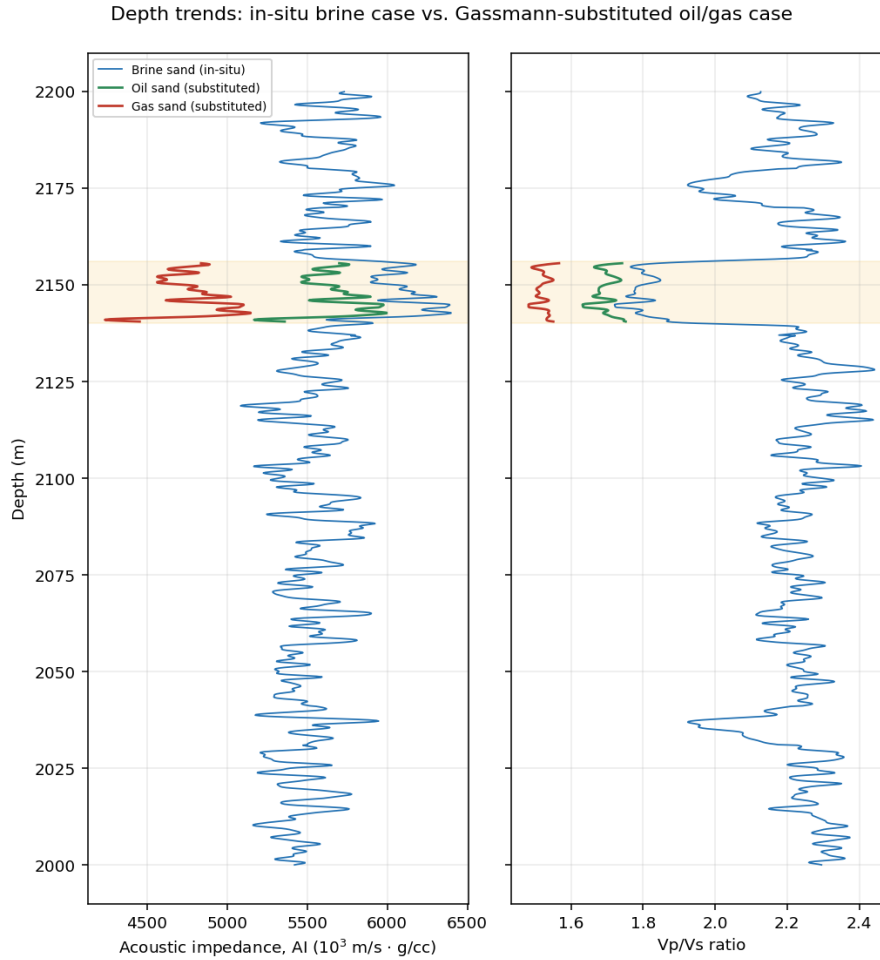


Figure 2: Depth trends of acoustic impedance and V_p/V_s ratio: in-situ brine case versus Gassmann-substituted oil and gas cases.

4.2 Fluid Substitution Results

Table 2 summarizes the average elastic properties of the clean reservoir interval (2142–2154 m, excluding the smoothed lithology-boundary transition zones) for each fluid case.

Relative to the in-situ brine case, gas substitution reduces V_p by 8.5%, bulk density by 14.0%, and V_p/V_s by 15.1% – consistent with the well-established rock-physics signature of gas: the pore-fluid bulk modulus collapses far more than the rock frame changes, so K_{sat} (and hence V_p) drops sharply while V_s , which depends only on the fluid-insensitive shear modulus and the

Table 2: Reservoir sand elastic properties by fluid case (clean-interval average).

Fluid	V_p (m/s)	V_s (m/s)	ρ (g/cc)	AI	V_p/V_s
Brine (in-situ)	2929	1637	2.093	6131	1.790
Oil (substituted)	2804	1660	2.036	5709	1.690
Gas (substituted)	2681	1765	1.800	4826	1.519

(comparatively modest) density change, is far less affected.

4.3 Rock Physics Template and Lithofluid Discrimination

Figure 3 shows the calibrated rock physics template. The three sand trends (brine, oil, gas) separate cleanly in V_p/V_s -AI space at fixed porosity, and the shale trend sits at systematically higher V_p/V_s for a given impedance – the classic basis for using this crossplot domain to discriminate both lithology and fluid type simultaneously. The well-log samples and synthetic core-plug measurements calibrate the reservoir-sand portion of the template well.

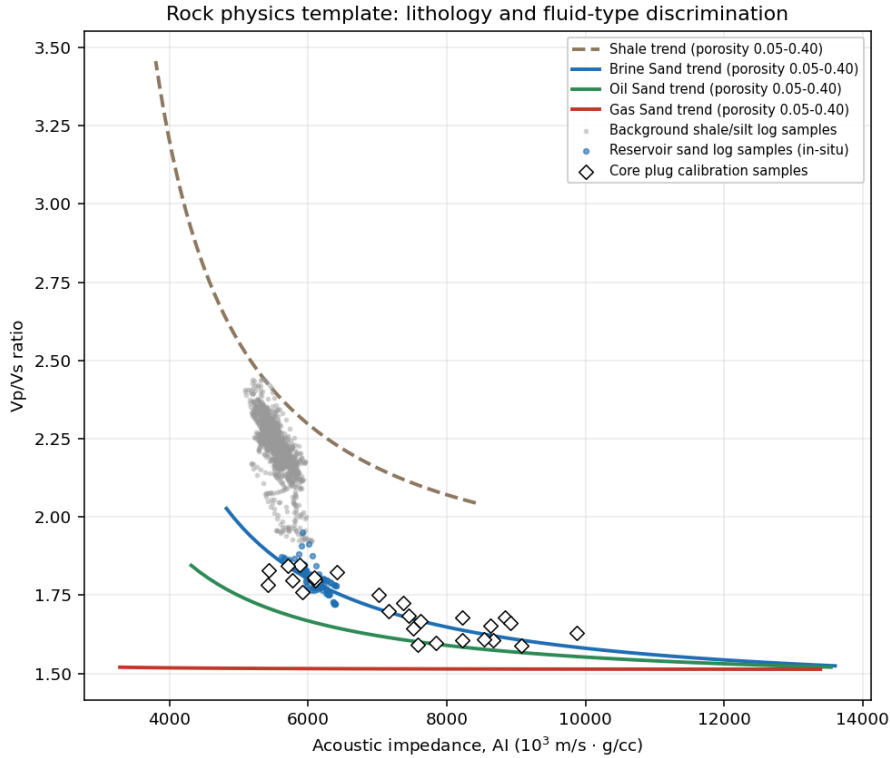


Figure 3: Rock physics template: V_p/V_s versus acoustic impedance, with brine/oil/gas-sand and shale trend lines, well-log samples, and synthetic core-plug calibration samples.

Figure 4 decomposes the same reservoir elastic states into the Lambda-Rho / Mu-Rho attribute domain (Goodway et al., 1997). Because Gassmann's formulation holds the shear modulus fluid-independent – and $\mu\rho \equiv \rho V_s^2 = \mu$ identically – the brine, oil, and gas points fall at essentially the same $\mu\rho$ while spreading widely in $\lambda\rho$ (incompressibility). This is a direct, first-principles illustration of why the LMR domain is used industry-wide for fluid discrimination: it isolates the fluid-sensitive term from the fluid-insensitive one. Shale, having different mineralogy and frame properties, separates cleanly on the $\mu\rho$ axis.

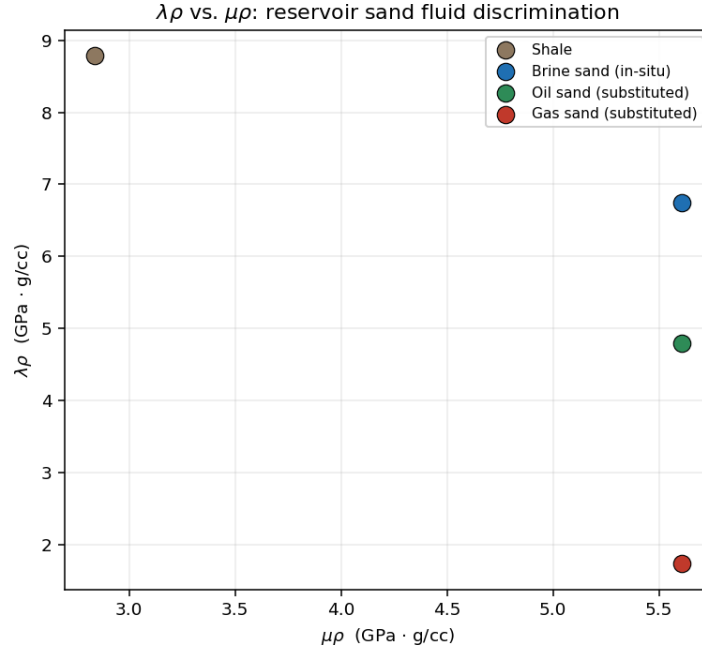


Figure 4: Lambda-Rho versus Mu-Rho: fluid discrimination in the reservoir sand, and lithology discrimination against the shale cap.

Figure 5 shows the V_p -porosity domain, in which the compaction trend and fluid-substitution trend are visually distinguishable: samples move along the porosity axis primarily due to compaction, and drop vertically (at roughly constant porosity) due to fluid substitution.

4.4 Synthetic AVO Gathers and Classification

Figure 6 shows the synthetic pre-stack angle gathers for the three fluid cases. The brine case shows a modest, single-polarity amplitude at the top-of-reservoir reflection; the oil case shows a markedly weaker near-offset response with amplitude building gradually with angle; the gas case shows a strong, clearly offset-dependent amplitude at both the top and base of the reservoir, with visible tuning interference between the two closely spaced (16 m) reflectors. Figure 7 presents the corresponding near/mid/far stacks, and Figure 8 the intercept-gradient classification.

Table 3 summarizes the resulting classification. The brine case ($A = +0.045$, $B = -0.280$) falls in Class I territory: the in-situ sand is slightly stiffer (higher impedance) than the shale cap, giving a positive intercept whose amplitude weakens with offset. Substituting oil moves the intercept to $A = +0.010$ – within a few hundredths of zero – while the gradient remains strongly negative; this is a genuinely subtle anomaly, straddling the Class I/Class II boundary, that would be difficult to identify from a near-offset stack alone and only becomes diagnostic once the far-offset amplitude and gradient are examined together. Gas substitution drives the response firmly into Class III territory ($A = -0.074$, $B = -0.390$): a classic unconsolidated-sand bright spot, with reflection amplitude growing markedly and unambiguously with offset.

5 Discussion

The workflow reproduces, from first principles, several well-established rock-physics and AVO relationships: the disproportionate sensitivity of V_p and V_p/V_s to gas relative to V_s ; the fluid-

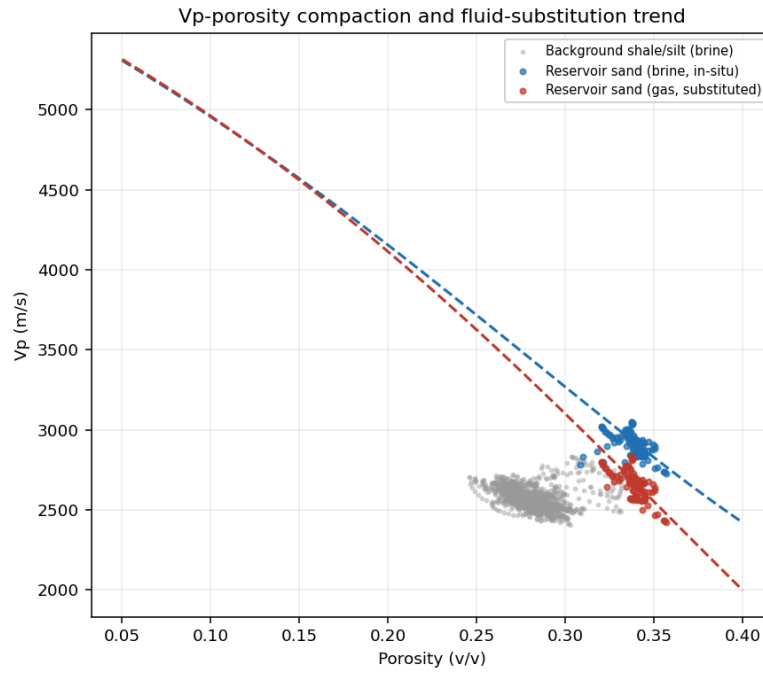


Figure 5: Vp-porosity crossplot showing the compaction trend (background shale/silt) and the fluid-substitution offset between brine and gas cases in the reservoir sand.

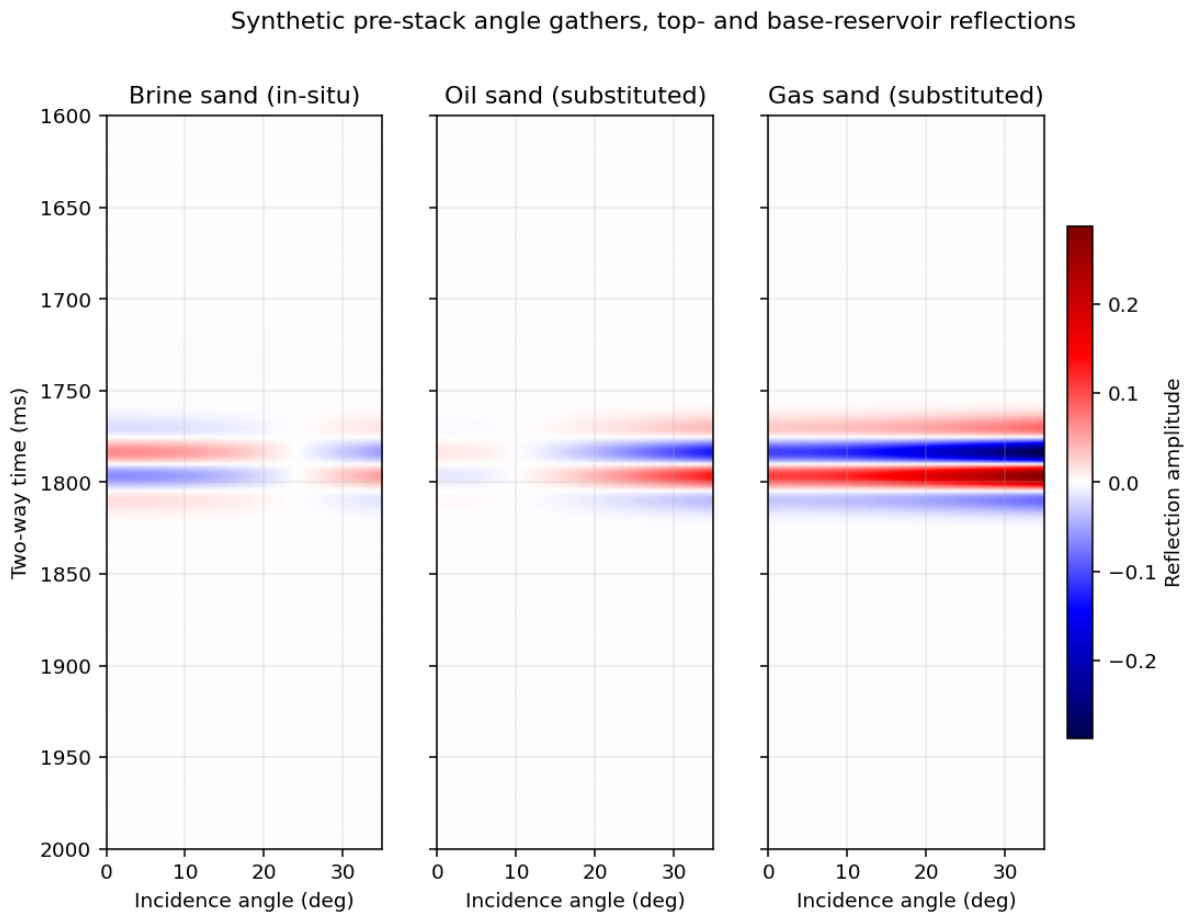


Figure 6: Synthetic pre-stack angle gathers (0–35°), top- and base-of-reservoir reflections, for the brine, oil, and gas cases.

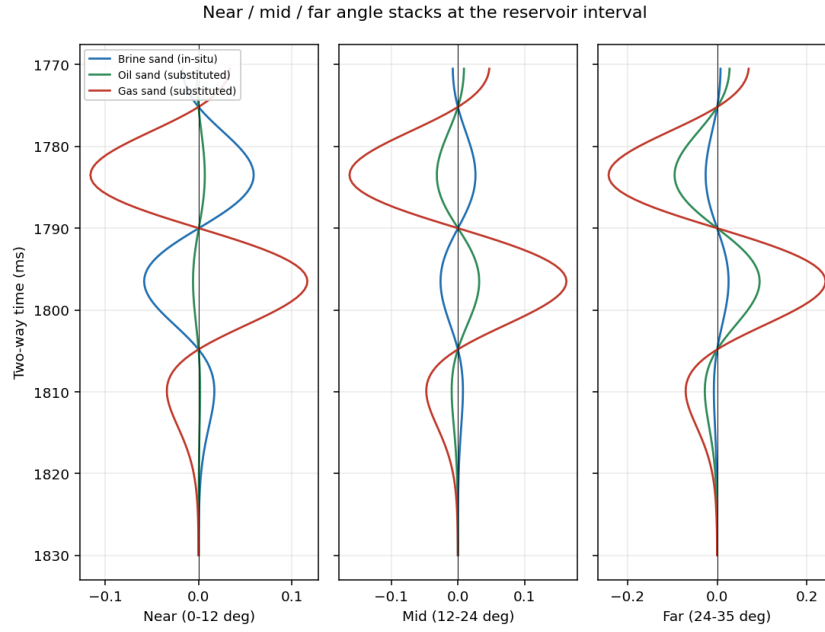


Figure 7: Near, mid, and far angle stacks at the reservoir interval, by fluid case.

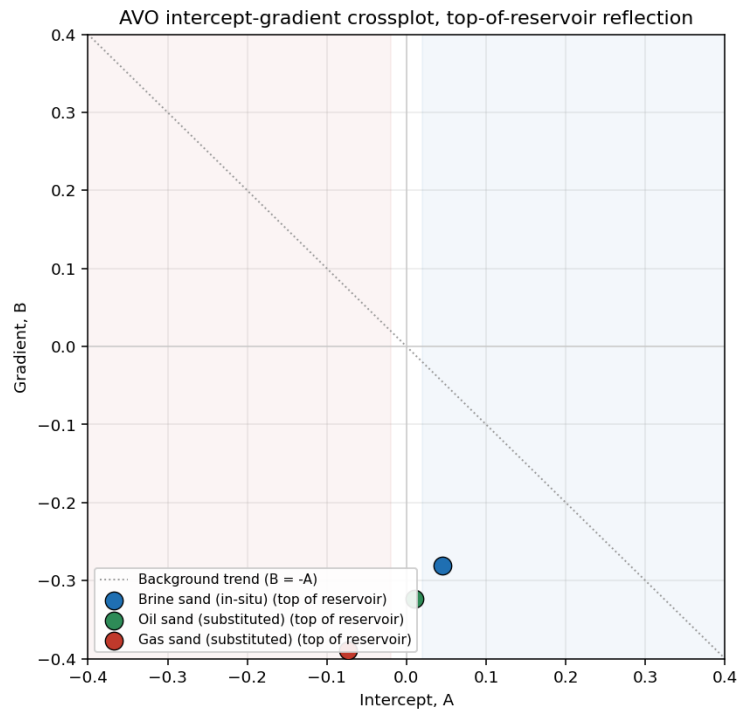


Figure 8: AVO intercept-gradient crossplot for the top-of-reservoir reflection, with Rutherford & Williams (1989) / Castagna & Swan (1997) class boundaries.

Table 3: AVO intercept/gradient classification, top-of-reservoir reflection.

Fluid	Intercept, A	Gradient, B	Classification
Brine (in-situ)	+0.045	−0.280	Class I
Oil (substituted)	+0.010	−0.324	Class II / IIp (subtle)
Gas (substituted)	−0.074	−0.390	Class III (bright spot)

independence of the shear modulus (and hence of $\mu\rho$) under Gassmann’s assumptions; and the progression from a Class I in-situ response, through a subtle, near-zero-intercept Class II/IIp anomaly for a moderate-impedance-contrast fluid case, to an unambiguous Class III bright spot for gas. The oil case is arguably the most instructive result here: it demonstrates concretely why AVO interpretation requires examining the full-angle gradient rather than a single stack, since its near-zero intercept would be easy to overlook without gradient information.

Several simplifications should be kept in mind when applying this workflow’s conclusions. The rock-physics model uses a uniform (Reuss-average, fully mixed) fluid saturation rather than a patchy-saturation model, which can meaningfully affect the seismic response at intermediate saturations. The oil and gas PVT properties are simplified representative relations rather than a full live-fluid Batzle-Wang treatment calibrated to a measured GOR and gas composition. The AVO modelling uses a linearized, weak-contrast reflectivity approximation valid to roughly 30–35°, and a 1D, normal-incidence-only wavelet convolution that does not capture offset-dependent tuning, NMO stretch, or anisotropy. None of these simplifications change the qualitative conclusions above, but a field application would need to revisit each of them against measured data.

6 Conclusions

- (1) A first-principles Gassmann fluid-substitution workflow, applied to a synthetic shaly-sand Tertiary reservoir, correctly reproduces the expected asymmetric sensitivity of V_p and V_p/V_s to pore fluid, with gas substitution producing an 8.5% reduction in V_p and a 15.1% reduction in V_p/V_s relative to the in-situ brine case.
- (2) A V_p/V_s -versus-acoustic-impedance rock physics template, calibrated against synthetic log and core data, cleanly separates brine, oil, and gas sand trends from the shale background, and a Lambda-Rho/Mu-Rho decomposition of the same data directly illustrates the physical basis for that separation.
- (3) Pre-stack AVO modelling of the three fluid cases classifies the in-situ (brine) response as Class I, the oil-substituted response as a subtle, near-zero-intercept Class II/IIp anomaly, and the gas-substituted response as an unambiguous Class III bright spot – demonstrating both a clear-cut and a genuinely difficult fluid-discrimination scenario within a single, internally consistent reservoir model.
- (4) The complete workflow, from synthetic log construction through fluid substitution, rock-physics templating, and pre-stack AVO modelling, is implemented in openly available Python source code and is fully and deterministically reproducible.

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